BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LARKANA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2017

PHYSICS

Past Paper 2017

Retyped faithfully from the original board paper — same questions, no changes.

Part –A: Multiple Choice Questions (MCQs Type A)

Time Allowed: 20 Minutes | **Maximum Marks:** 16

Note: Instruction: Identify the correct answer of each (MCQ) from its given four options and write it on the answer script/copy. Dated: 29 April 2017. Objective Types B and C contain the same 16 questions in a different order.

Q1. The dimensional formula for spring constant (k) is.....

(A) ML^0T^{-2}
(B) $M^2L^0T^{-2}$
(C) $M^{-2}L^0T^{-2}$
(D) None

Q2. All the non-luminous objects are visible due to the.....of light from their surfaces.

(A) Regular reflection
(B) Irregular reflection
(C) Both
(D) None

Q3. A body is dropped from a height of 4m , its velocity just before hitting the ground is.....

(A) 8.85 cm /sec
(B) 8.85 m/sec
(C) 88.5cm/sec
(D) None

Q4. In uniform circular motion a body posses only.....

(A) Angular acceleration
(B) Centripetal acceleration
(C) Tangential acceleration
(D) None

Q5. The speed of sound is greater in solids than gases due to their high.....

(A) Density
(B) Pressure
(C) Elasticity
(D) None

Q6. The velocity of sound is the greatest in.....

(A) Water
(B) Air
(C) Steel
(D) None

Q7. Artificial gravity is produced by spinning the space ship.....

(A) Along its own axis
(B) Around its own axis
(C) Both
(D) None

Q8. Half horse power =.....

(A) 373 watt
(B) 373 k watt
(C) 746 J
(D) 746 watts

Q9. If the length of simple pendulum is doubled, its time period is...

(A) 2 T
(B) 1.5T
(C) 1.414 T
(D) None

Q10. The distance between the objective and eye-piece is called...

(A) Object distance
(B) Image distance
(C) Focal Length
(D) None

Q11. Five equal forces of 10 N each are applied at one point and all are lying in one plane. If the angles between them are equal the resultant force will be.....

(A) Zero
(B) 10 N
(C) 50 N
(D) None

Q12. A weight varies with.....

(A) Location
(B) Value
(C) Both
(D) None

Q13. The eye piece of compound micro scope is called as...

(A) Diffraction grating
(B) Magnifying glass
(C) Prism
(D) Telescope

Q14. If the mass of earth becomes four times the value of g becomes .....times.

(A) Two
(B) Four
(C) Eight
(D) None

Q15. The time taken by the moon to complete one rotation around earth is.....

(A) 365 days
(B) 7 days
(C) 27.3 days
(D) None

Q16. If given spring constant K is cut into three identical segments the spring constant of each segment is.....

(A) K/2 (B) K/3

(C) 3 K (D) None

Part –B:- Constructed Response Questions(CRQs)/ (Short Answer Questions) (45 Marks)

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 69

> Note: Attempt any NINE of the following questions. Dated: 29 April 2017. Maximum marks 69 is the total for Part B and Part C together as printed on the subjective sheet.

Q1. For what purpose, the sine law and cosine law are used?

Q2. How would you locate the centre of gravity of an irregular piece of a metal sheet?

Q3. Why does the speed of sound wave change with temperature?

Q4. Mention some illustrations of importance of resonance in musical instruments?

Q5. Two forces of magnitude 5 N and 10 N act on a body in directions making angles 30° and 60° respectively with x-axis. Find the resultant force.

Q6. Why the light rays could pass through the vacuum but not sound waves?

Q7. In which condition the standing waves produce the vibrational frequency of 2nd over tone.

Q8. If one of the rectangular components of vector is not zero. Can its magnitude be zero?

Q9. Can interference of visible light produce dark and bright fringes?

Q10. Why distant objects look small in size?

Q11. Why would it be advantageous to use blue light with compound micro scope?

Part –C:- (Extended Response Questions(ERQs)/(Long Answer Questions) (24 Marks)

> Note: Attempt any TWO of the following questions.

Q12.

(a) What do you know about Dot product and Cross product? Discuss in detail?

(b) A body of mass 3 kg is thrown up vertically K.E of 450 Joules. Find the height If K.E of the body becomes half its original value is given.

Q13.

(a) Describe Youngs' double slit experiment. What are the conditions for maxima and minima?

(b) The distance between the slits in Young's double experiment is 0.22 cm interference fringes are formed on screen at a distance of 100 cm from the slits. The distance the third dark fringe from the central bright fringe is 0.052 cm. Find the wave length of the incident of light.

Q14.

(a) What is converging lens? Derive thin lens formula by showing the ray diagram.

(b) The line action of force F passes through a point P of a body whose position vectors in meter is i+2J+K F=2i-3J+4k(in Newton. Determine the torque about the point A whose position vector(in meter) is 2i+J+k.